

John David Prieto Prada

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Citizenship: Colombian

Research interests

Reinforcement learning, deep learning, robot manipulation, human-robot interaction, virtual reality, augmented reality, and mixed reality.

Education

2018 – 2024 **Daegu Gyeongbuk Institute of Science and Technology (DGIST)** – Daegu, South Korea
PhD in Robotics and Mechatronics Engineering
Supervisor: Professor Cheol Song. *Intelligent Bio-OptoMechatronics lab.*

2016 – 2018 **Daegu Gyeongbuk Institute of Science and Technology (DGIST)** – Daegu, South Korea
M.S. in Robotics Engineering
Supervisor: Professor Cheol Song. *Intelligent Bio-OptoMechatronics lab.*

2010 – 2015 **New Granada Military University** – Bogota, Colombia
BA in Mechatronics Engineering
Supervisor: Professor Alvaro Joffre Uribe Quevedo.

Honors

2017 DGIST excellent student researcher award (Daegu Gyeongbuk Institute of Science and Technology).

Publications

2025 **Group-Orbit Exploration: Symmetry-Aware Reinforcement Learning for Push-and-Grasp Robotic Manipulation**
John David Prieto Prada, Hogun Ha, Jun Hyeok Lee, and Hyunki Lee.
IEEE Transactions on Pattern Analysis and Machine Intelligence, under review.

2025 **Sim2Real2Sim Push-and-Grasp: A Reinforcement Learning Framework with Optical Tracking Feedback**
John David Prieto Prada, Hogun Ha, Changmin Lee, and Hyunki Lee.
IEEE Robotics and Automation Letters, under review.

2023 **A Gaze-Speech System in Mixed Reality for Human-Robot Interaction**
John David Prieto Prada, Myung Ho Lee, and Cheol Song.
IEEE International Conference on Robotics and Automation (ICRA).

2022 **A Deep Learning Technique as a Sensor Fusion for Enhancing the Position in a Virtual Reality Micro-Environment**
John David Prieto Prada, Miguel Luna, Sang Hyun Park, and Cheol Song.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

2021 **Enhanced location tracking in sensor fusion-assisted virtual reality micro-manipulation environments**
John David Prieto Prada, Jintaek Im, Hyondong Oh, and Cheol Song.
PLoS One.

2017 **Exploring the Potential of Modifying Visual Stimuli in Virtual Reality to Reduce Hand Tremor in Micromanipulation Tasks**
John David Prieto Prada, Taiwoo Park, Sunjin Jang, Jintaek Im, and Cheol Song.
Current Optics and Photonics.

Research experience

Sep 2026 – present
InnoCore Post-Doc researcher at CCG Lab
Supervisor: Professor HyunKi Lee (DGIST).

- Develop Medical Physical AI methods for adaptive and safe robotic assistance, including reinforcement learning, multimodal perception, shared autonomy, and sim-to-real validation. tasks.
- Implement reinforcement learning, multimodal perception, shared autonomy, and sim-to-real validation for medical robots. tasks.

Sep 2024 – Aug 2026
Post-Doc researcher at CCG Lab
Supervisor: Professor HyunKi Lee (DGIST).

- Building a common platform for logistics service robots.
- Implementation of a simulation environment for logistic service robots.
- Implementation of machine learning algorithms for robot learning in different robotic tasks.

Sep 2018 – Aug 2024
Doctoral research student at IBOM lab
Supervisor: Professor Cheol Song (DGIST).

- Implementation of reinforcement learning techniques for robotic manipulations in mixed reality environments.
- Robotic teleoperation using virtual, augmented and mixed reality technologies.
- Implementation of machine learning algorithms for virtual reality tracking in small-scale applications.

Sep 2016 – Aug 2018
Master research student at IBOM lab
Supervisor: Professor Cheol Song (DGIST).

- Implementation of sensor fusion techniques for virtual reality tracking in small-scale applications.
- Virtual reality simulations for entertainment.

Teaching experience

Spring 2025 **Lecturer at DGIST, Intelligent Robotics (IE803)**

- Fundamentals in robot kinematics.
- Design of robotic simulations for training.
- Design of machine learning techniques.

Average student rating: 4.5/5.

Summer 2023 **Teacher at DGIST, Pre-Undergraduate Research Participation Program: Reinforcement learning for robot control**

- Fundamentals in reinforcement learning.
- Design of robotic simulations for training.
- Design of intelligent agents for robot control.

Average student rating: 4.5/5.

Summer 2022 **Teacher at DGIST, Pre-Undergraduate Research Participation Program: Implementation of a robotic teleoperation in mixed reality environments**

- Fundamentals in robotic manipulators.
- Fundamentals in mixed reality devices.
- Design of virtual interfaces for robot control.
- Deploying mixed reality application for real-world scenarios.

Average student rating: 4.5/5.

Summer 2020 **Teacher at DGIST, Pre-Undergraduate Research Participation Program: Fundamentals in C# language for virtual reality gaming**

- Fundamentals in C# language.
- Fundamentals in Unity software.
- Design of basic virtual objects.

Average student rating: 4.5/5.

Technical skills

Programming languages
Proficient in: C#, Python and Latex.
Familiar with: ROS, ROS 2, C/C++ and Matlab.

Software
Unity, Vuforia, Blender, Solidworks and Arduino.

Frameworks
Pytorch and Tensorflow.

Languages
Spanish (native), English (fluent), Korean (basic)